Dendrochronology

Reading Environment and History through Tree Rings

An introductory course · Lesson 1

Hitoshi Yonenobu

An introduction with no advance preparation

We will introduce the terms and read the figures together.

Purpose	How tree rings help us understand environment and history.
Dates	Which event a measured age represents.
Methods	A suitable method for a sample, and the conditions to check.

Dates help us frame better questions

When did it happen?

Place building work, forest change and climate variation on a timeline.

Which came first?

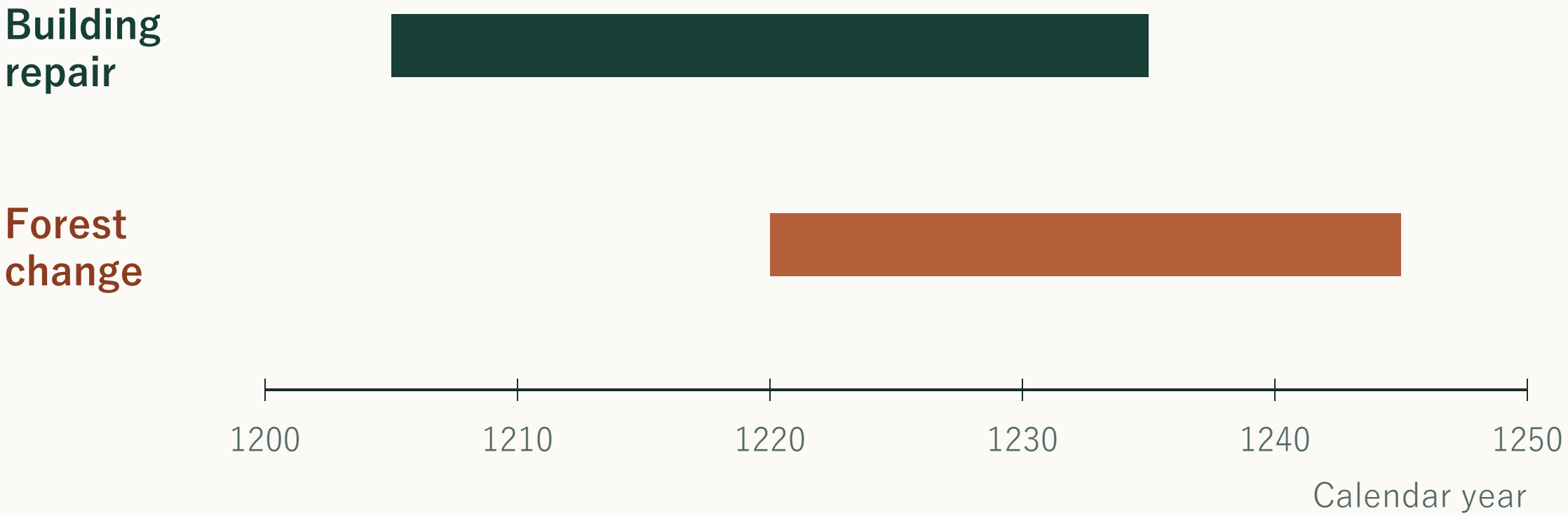
Use order and time intervals to assess possible explanations.

What can we conclude?

Events occurring together require further evidence of causation.

Can overlapping age ranges establish order?

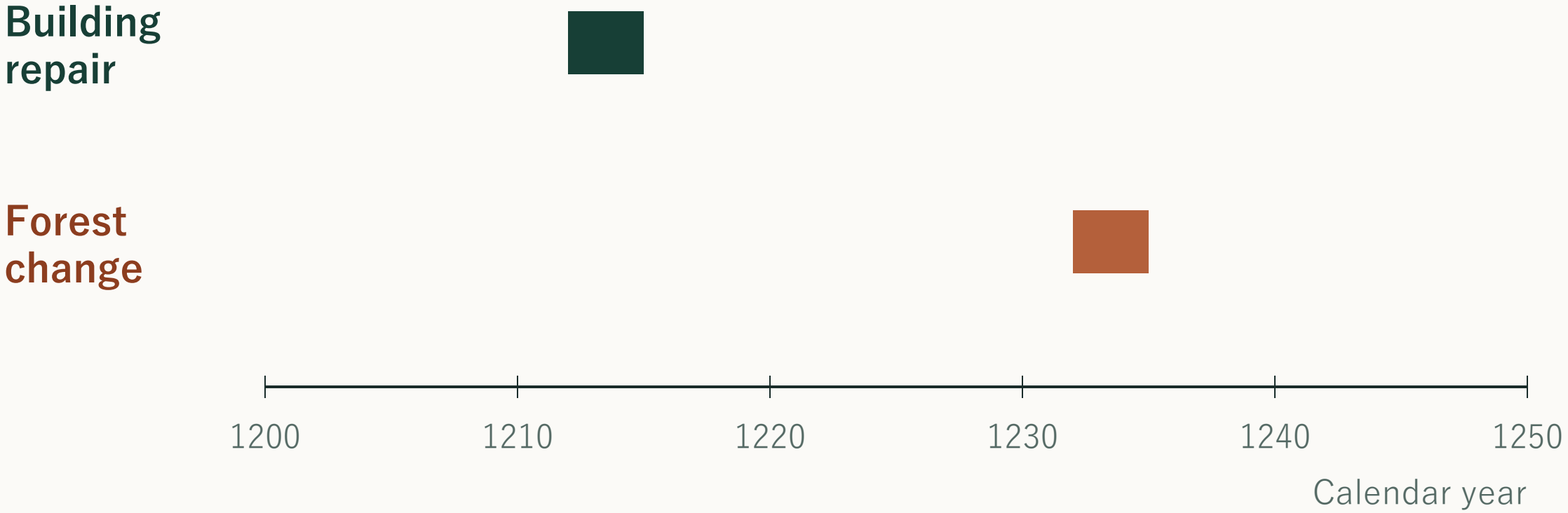
Overlapping ranges may leave the order of events unresolved.



Illustrative ranges without a specified confidence level.

Establishing order leads to the next question

Here, repairs came first. Explaining the cause requires other evidence.



Illustrative ranges without a specified confidence level.

What can real wood samples tell us?



Buried wood

Investigate the timing of forest or landscape change.



Excavated wood

Investigate a site and its history of wood use.



Historic timber

Investigate construction, repair and reuse.

Photos from Yonenobu's 2021 lecture, slide 30. Context connects samples to events.

Relative and numerical ages

Relative age

Places events in order

Example: a lower and an upper layer

Check for overturning or redeposition.

Numerical age

Expresses age as a number or range

Example: a ring formed in 1300 CE

Absolute dating still has uncertainty.

Does the age of wood date its use?

If old buried wood becomes modern furniture, what does its date represent?

Ring formation
1200 CE

→

Felling
1300 CE

→

Construction
1320 CE

→

Reuse
1450 CE

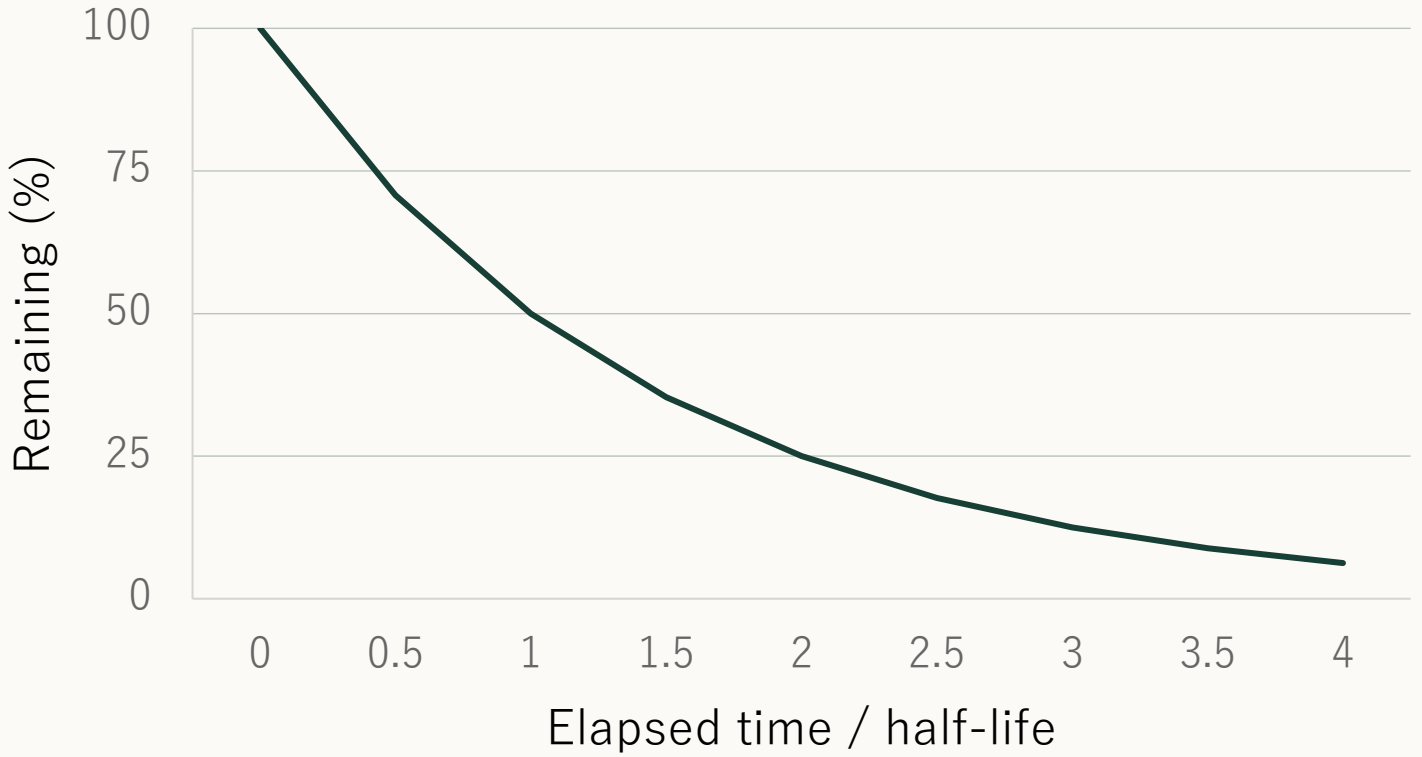
Ring formation, felling, construction and reuse are separate events.

Schematic. Intervals are not drawn to scale.

Age, sample relationships and dating clocks

Perspective	Example	Question to check
Relative age	Stratigraphy / typology	Which event came first?
Numerical / absolute	Tree rings / radiocarbon	What age or range?
Direct / indirect	Object / related deposits	How does the sample relate?
Record / clock	Growth / decay / signals	What starts and preserves it?

Radioactive decay and half-life



100%
50%
25%
12.5%

After 0, 1, 2 and 3 half-lives

For a large population, each half-life halves the proportion remaining.

What event can a charred seed help date?

What we measure

Carbon-14 in material derived from living organisms.

Connecting the events

Check how plant growth relates to activity at the site.

Relating the result to calendar time

Use a calibration curve to assess a calendar-age range.



Charred maize from Ceibal

Photo from Yonenobu's 2021 lecture, slide 92

Dating mineral cooling and carbonate growth

K–Ar / Ar–Ar

Examine suitable volcanic minerals.

Assess timing related to cooling and gas retention.

For ash, check the minerals and depositional context.

U–Th

Examine suitable carbonates, including stalagmites.

Use isotopic changes following carbonate formation.

Assess initial material and later chemical exchange.

Dating sand burial and the cooling of a kiln

Luminescence dating

Measure stored signals produced by radiation.

During measurement, OSL uses light and TL uses heat.

Check the starting exposure or heating and signal resetting.

Archaeomagnetic dating

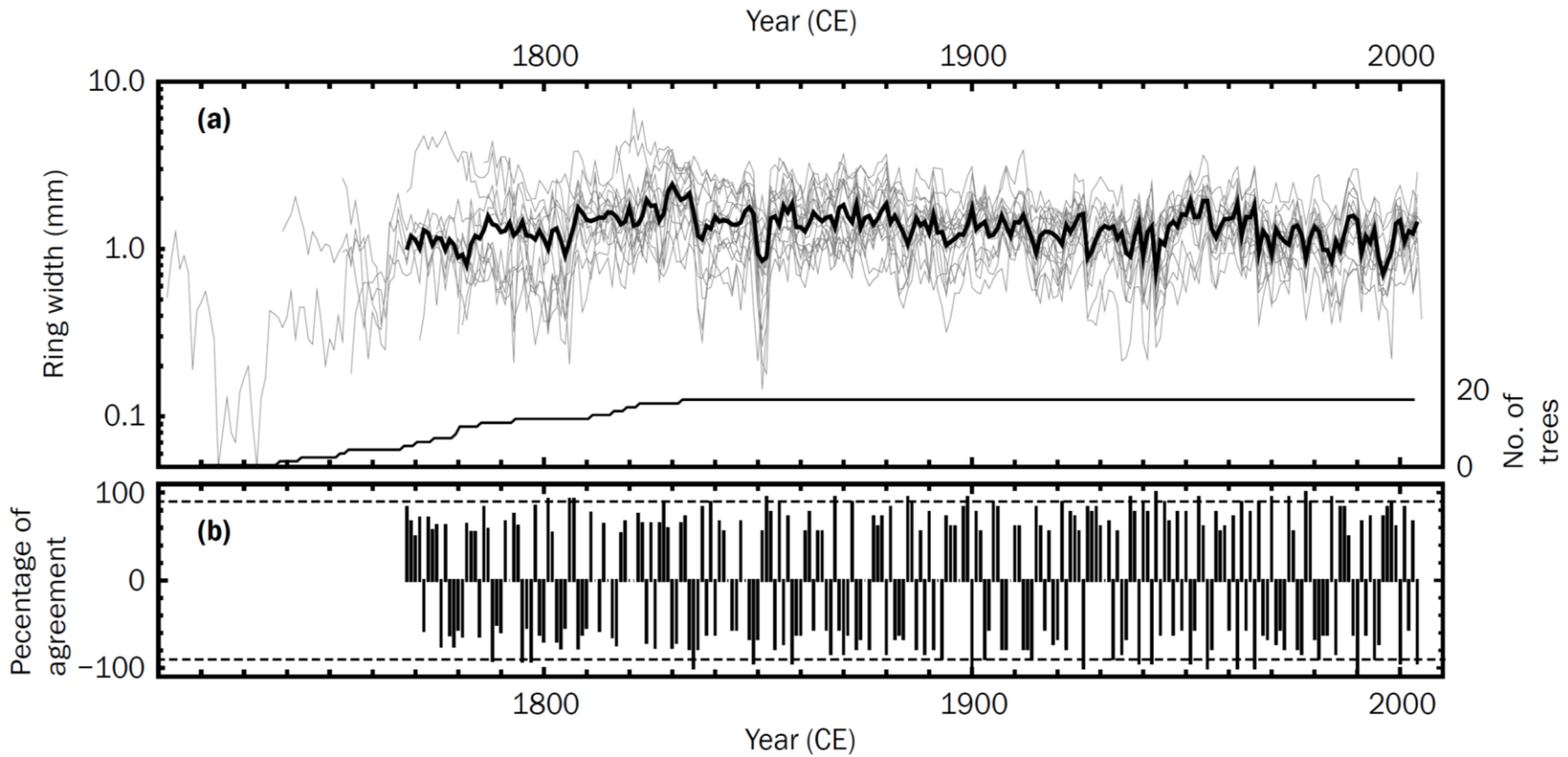
Examine magnetism preserved in fired materials.

Compare the record of final cooling after sufficient heating with a regional curve.

Check for movement after firing and record the location.

Annual patterns in measured ring widths

Find years when several series narrow together.



Grey lines

Individual series

Thick black line

Combined chronology

Lower panel

Agreement in change

Yanase sugi, 1768–2004 CE: measurements from 18 discs. Ring width uses a logarithmic axis.

Yanase sugi: original paper

Learning crossdating

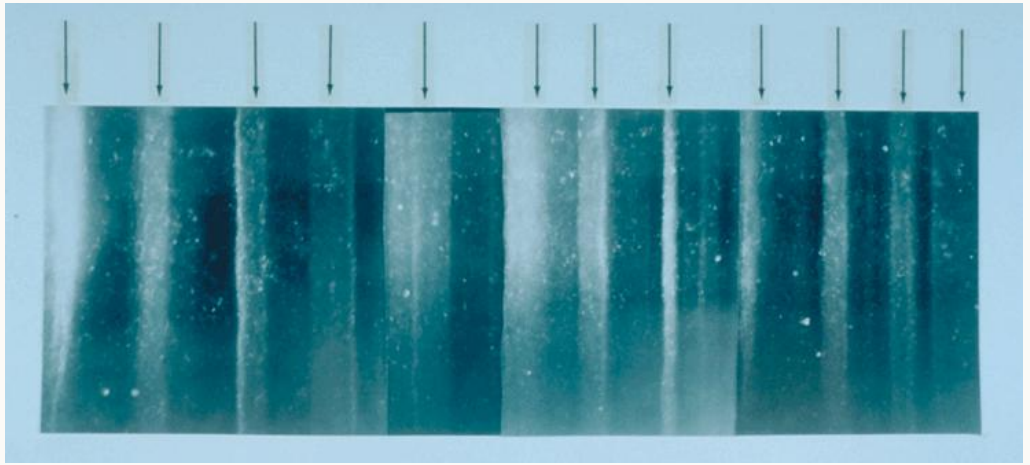
Layers in wood, sediment and ice

Compare what is recorded and how its age is established.



Lake sediment: Ichinomegata

Study layering, pollen and chemical composition.



Ice core: Greenland

Ice isotopes and trapped gases record climate and the atmosphere.

Assess resolution, continuity and dating for each archive.

Left: cropped photo from Yonenobu (2021), slide 69. Right: NOAA / USACE, public domain.

Source of the ice-core image

Environmental proxy records

What could explain a narrow ring?

Compare with observations

Identify relationships with seasonal temperature or moisture.

Consider other influences

Age, competition and insect damage can also affect ring width.

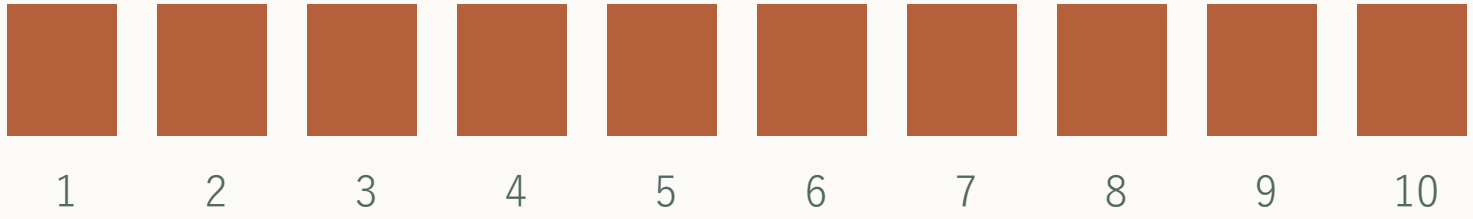
Check where the relationship applies

Test transfer to other species, regions or periods.

Annual resolution and calendar placement

Distinguish the interval between observations from their position in time.

Annual steps
Unplaced in time



Annual steps
Calendar dated



Schematic. Fine resolution and calendar placement are different properties.

Different records on a common time scale

Check the dating scale

Check calendar versus radiocarbon ages, reference years and uncertainty.

Compare the same phenomenon

Spring temperature and summer drought describe different conditions.

Investigate disagreements

Consider dating, response lags, location and preservation.

Sample A: a worked example

We will work through A together, then practise with B–D.

Measurement	Radiocarbon in the charred seed.
Dated event	When the plant incorporated carbon.
Further check	Does this seed relate to activity at this hearth?

Four samples and possible dating methods

ID	What is available?	Target event
A	A charred seed from a hearth	Activity involving the fire
B	Historic timber with outer wood removed	Construction or repair
C	Fired material in an apparently undisturbed kiln	The last firing
D	Wind-blown sand buried at the site	Sand deposition

Connecting measurements to events

Sample and clock

Which record can we use: carbon, rings, magnetism or a stored signal?

Meaning of the age

Does it date formation, cooling, burial, felling or use?

Further checks

Check contamination, missing outer wood, movement, resetting and context.

Candidate methods and further checks

ID	Possible method	Meaning and checks
A	Radiocarbon dating	Dates the seed’s carbon. Check its link to hearth use.
B	Dendrochronology	Dates surviving rings. Assess outer-wood loss and reuse.
C	Archaeomagnetism	Records cooling. Check the regional curve and later movement.
D	OSL	Time since burial after exposure to light. Check resetting.

Age notation and reference years

1000 ± 30 BP

Example of a radiocarbon age, referenced to 1950.

Subtracting it from 1950 does not give a calendar age.

cal BP / cal BCE / cal CE

Calibrated calendar ages. Record the curve and assumptions.

The reference year for cal BP is also 1950.

1300 CE

Example of a ring's calendar year of formation.

Historical CE / BCE notation has no year zero.

Today’s questions, in your own words

Which event do you want to date?

Choose a sample and method, then explain the link to the event.

What does the sample record?

Establish the date before interpreting environmental information.

What can the evidence support?

Distinguish conclusions supported by the date from those needing other evidence.

After-class review with figures and sources

Useful keywords

dendrochronology

crossdating

Focus the search

dendrochronology crossdating tutorial

radiocarbon calibration sample context

Start with the caption

Identify the sample, region and period shown.

Follow the references or DOI to the original paper.

Original sources and open data for review

Overview of dendrochronology	Wikipedia Concepts and research examples
Measured Yanase sugi chronology	Yonenobu, Hoshino & Ohyama (2021) Original paper
Samples and radiocarbon calibration	University of Oxford Methods and interpretation
Environmental archives and proxies	NOAA Resources and explanations
International Tree-Ring Data Bank	NOAA NCEI Open data
Image sources and licences	Wikimedia Commons Information for each image

Next lesson: wood anatomy and tree-ring structure

Where is the cambium, and what does it produce?

How do earlywood and latewood differ?

How can we identify an annual boundary?



An example of a wood cross-section

Photo: Arnoldius / CC BY-SA 2.5

Source of the wood image

Image licence